Fiscal Year:	FY 2011	**Task Last Updated:**	FY 03/08/2011
PI Name:	Weitz, David Ph.D.		
Project Title:	Physics of Colloids in Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Fluid Physics Rapid Turn Around Flight Opportunity
Start Date:	12/18/2007	**End Date:**	12/17/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: per W. Meyer/GRC, title should be "Physics of Colloids in Space"; also, change in period of performance--should be 12/18/2007-12/17/2012 for first awarded period of performance and supplements; there was no end date change to 3/31/2013 (Ed., 4/21/15) NOTE: title change to "Binary Colloidal Alloy Test -6" and change in end date to 3/31/2013, per W. Meyer/GRC (Ed., 9/8/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yodh, Arjun (University of Pennsylvania) Chaikin, Paul (New York University) Lynch, Matthew (Proctor & Gamble Company) Hollingsworth, Andrew (New York University) Lu, Peter (Harvard University) Frisken, Barbara (Simon Fraser University) Bailey, Arthur (Simon Fraser University)		
Grant/Contract No.:	NNX08AE09G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The BCAT-6 and BCAT-7 investigations build on a heritage of (3 ISS and 2 Shuttle/Mir) flight experiments and they are packed with important scientific and commercial investigations. This effort will enable us to continue to address fundamental questions in colloidal engineering that impact product shelf life and answer how concentrated systems of particles of select sizes and shapes cause order to naturally arise out of disorder when gravity is removed. The baseline proposal is to deliver BCAT-6 for Multi-Purpose Logistics Module (MPLM) stowage on ULF-5 presently planned for 7/29/10. BCAT-7 is planned for late stowage on ULF-5. We propose to conduct four parallel sets of sample monitoring and imaging experiments (20 samples in all): (1) Crystallization of (NYU-provided) ellipsoidal colloidal particles and PS-DNA particles that should form crystals. The PS-DNA tagged particles will be set to crystallize at room (ISS cabin) temperature and be annealed as the ISS cabin temperature cycles a degree or two each side of this whenever the International Space Station (ISS) orbits the earth and is hidden from the sun when it goes behind the earth. PS-DNA also forms the foundational blocks for making future self-replicating structures, (2) Equilibration of (UPenn-provided) graphene disks with liquid crystal properties that should be visible between crossed-polarizers as they approach equilibrium after initially being homogenized (looking at a new generation of highly sensitive detectors and the foundations for zero-resistance substrates based on carbon electronics), (3) Phase separation of colloidal (P&G- and Harvard-provided) samples of commercial interest in a multi-billion dollar industry where the physics underlying shelf-life is masked by gravitation, and (4) the Competition between phase separation and crystallization (with SFU providing samples); this is the cause of defects that are often seen during the extrusion of plastics and a better understanding of the underlying physics should result from additional data points in this investigation. The objectives of these experiments examine different conditions that result in entropically-driven colloidal crystallization, melting, self-organization, and phase separation of systems. The long-duration low-gravity environment provided on the International Space Station (ISS) allows such structures to form and persist for sufficient time to establish the evolution toward equilibrium through time sequenced images taken with the EarthKAM system on the ISS or with the intervalometer of the new camera (until the EarthKAM software is updated to communicate with it). These experiments cannot be done on Earth because of problems caused by sedimentation, convection, and jamming. Being able to track the time evolution allows scientists to see the underlying physics, which is of particular relevance to product manufacturers for the pharmaceutical, food, and cleaning industries (in terms of shelf life and mechanical and thermodynamic properties). Of particular note is to have the first low-gravity flight of self-annealing PS-DNA tagged particles, ellipsoidal colloidal particles, and 2-D graphene carbon sheets in equilibrium. The fundamental questions outlined above can be addressed using two Binary Colloid Alloy Test (BCAT) flight hardware modules that already exist; and the answers provided by these experiments will lay the foundations for later scientific investigations proposed by these investigators for the Advanced Colloids Experiment (ACE) with the Light Microscopy Module (LMM) on the ISS. Funding this proposal accomplishes two objectives. First the direct scientific results will be important on their own and guide the choice of which classes of questions can be resolved in microgravity while telling us what needs to be experimentally tweaked now to get things to work with the LMM (i.e., particle size, polydispersity, concentration, etc.); and second, this work will form a successful bridge for keeping the science teams of existing and future students alive and productive while working with our experienced scientists (PIs) to ensure that these endeavors are optimized and focused while we continue the development of and await the future deployment of the fully functional Light Microscopy Module on the ISS in 2013 or 2014.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made significant progress toward a number of goals associated with the colloids effort in microgravity in 2010. We have continued analyzing important data from ISS-based astronauts as part of the Binary Colloid Alloy Test (BCAT) series of experiments. We have prepared and delivered samples as part of the BCAT-6 experiment which are scheduled to be launched on Space Shuttle Discovery later this year. We have begun preliminary development of samples, preparation and analysis protocols, and apparatus for upcoming BCAT-7, PACE and ACE experiments. We have made a number of our findings public through publication in peer-reviewed journals, and conference presentations. We published the first paper with results from earlier iterations of the BCAT experiments, in the Journal of Real-Time Image Processing, documenting some new methods we are using to process the unique images from ISS. We have presented the BCAT results to the AIAA conference, among the most important aerospace meetings in the country.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Lu PJ, Hidekazu O, Frey C, Chamitoff GE, Chiao L, Fincke EM, Foale CM, Magnus SH, McArthur Jr WS, Tani DM, Whitson PA, Williams JN, Meyer WV, Sicker RJ, Au BJ, Christiansen M, Schofield AB, Weitz DA. "Orders-of-magnitude performance increases in GPU-accelerated correlation of images from the International Space Station." Journal of Real-Time Image Processing 2010 Sep;5(3):179-93. http://dx.doi.org/10.1007/s11554-009-0133-1 , Sep-2010